Characterizing Filter Phase Response

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Key Concepts

1) The phase of a complex number (e.g., filter phase response) is not uniquely defined since any multiple of 2π radians can be added without changing the value of the complex number.

    a) The principal value of phase is limited to the range $-\pi < \text{ARG}\{H(e^{j\omega})\} \leq \pi$.

    b) Unwrapped phase is denoted as $\arg\{H(e^{j\omega})\}$ and is obtained by adding multiples of 2π to the phase to make it as continuous a function of ω as possible.

2) Linear phase filters ($\arg\{H(e^{j\omega})\} = -\omega n_0$) only introduce a time delay (n_0) and do not cause any distortion.

3) Generalized linear phase filters can have discontinuities of π in the phase response corresponding to sign changes in the frequency response (locations where $H(e^{j\omega})$ crosses zero).

4) Group delay is defined as the negative derivative (or slope) of the phase at a particular frequency.

    a) Group delay is the number of samples a sinusoid at that frequency is delayed.

b) Constant group delay implies linear phase. All input frequencies experience the same delay and thus there is no distortion.

c) If the group delay is not constant, then different frequency components of the input experience different delays through the system. This changes the alignment of the frequency components of the output and thus changes the "shape" of the input signal.

Video Screenshots

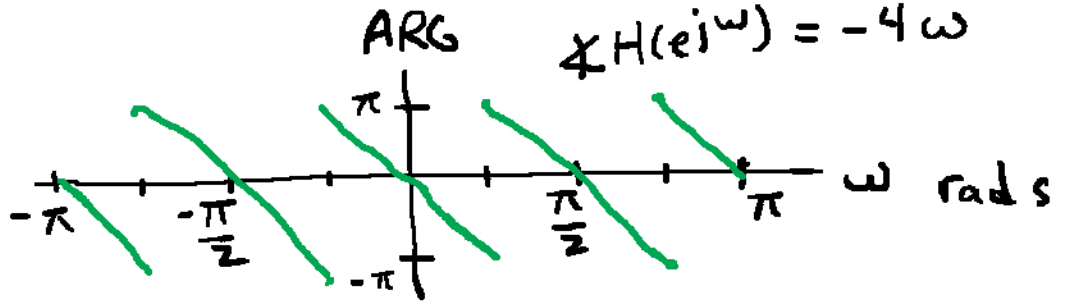
Frequency response: $H(e^{j\omega}) = |H(e^{j\omega})|\, e^{j\angle H(e^{j\omega})}$

1) $\angle H(e^{j\omega})$ is not uniquely defined

$$e^{j\angle H(e^{j\omega})} = e^{j(\angle H(e^{j\omega}) + \ell 2\pi)}$$

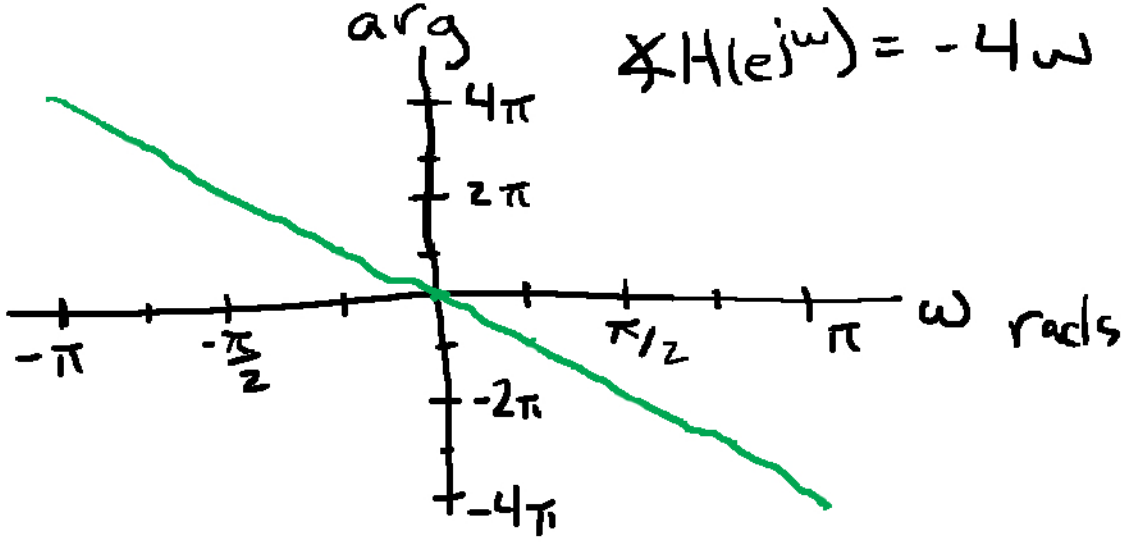
2) Principal value

$$-\pi < \text{ARG}\{H(e^{j\omega})\} < \pi$$



3) Unwrapped phase

$\arg\{H(e^{j\omega})\}$



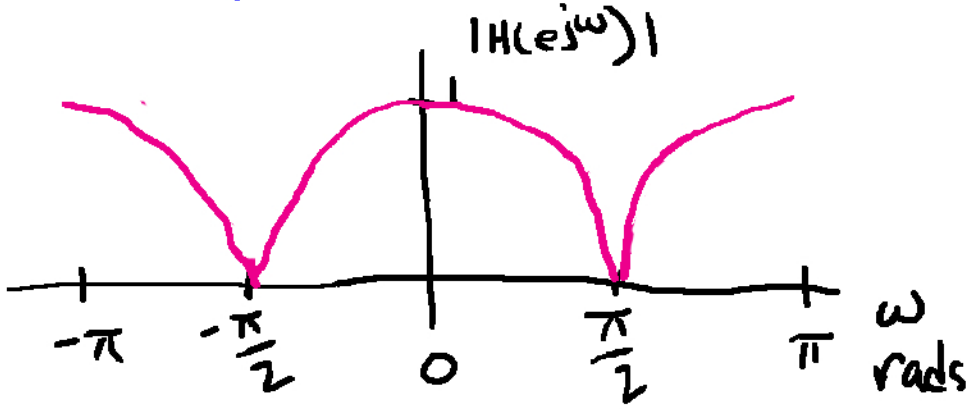
- Linear phase: $\arg\{H(e^{j\omega})\} = -\omega n_0$

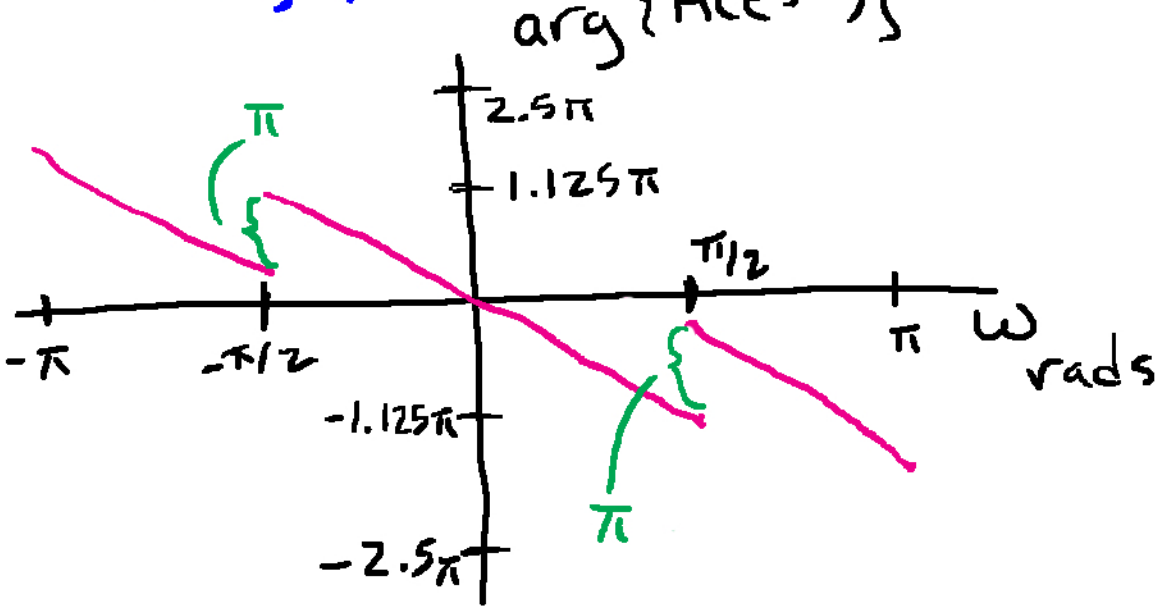
  no distortion other than a time delay

- Generalized linear phase: $H(e^{j\omega}) = A(\omega)\, e^{-j\alpha\omega + j\beta}$

  $A(\omega)$ is real, but can be negative

Example: $A(\omega) = \cos(\omega)$, $\alpha = 2.5$, $\beta = 0$

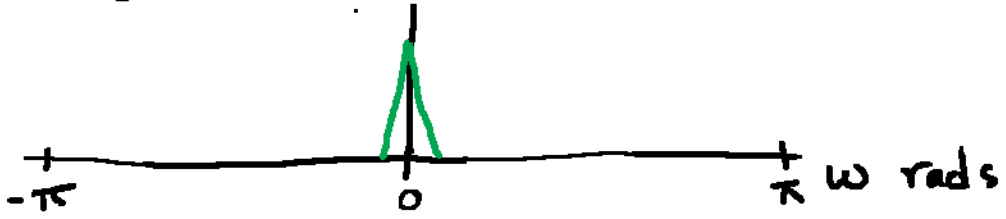




- Nonlinear phase:

Group delay - characterize nonlinear phase

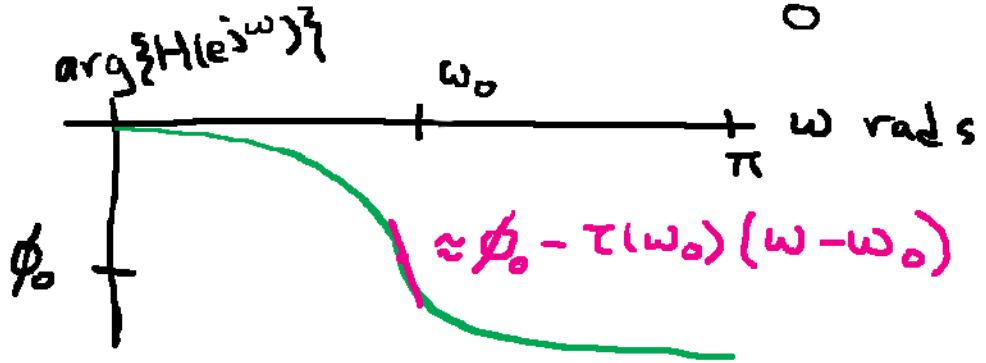
$s[n] \xleftrightarrow{DTFT} S(e^{j\omega})$



$x[n] = s[n]e^{j\omega_0 n} \xleftrightarrow{DTFT} X(e^{j\omega}) = S(e^{j(\omega-\omega_0)})$



$|H(e^{j\omega})| = 1$



$$\tau(\omega) = -\frac{d}{d\omega}\arg\{H(e^{j\omega})\}$$

group delay

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega}) = S(e^{j(\omega-\omega_0)})e^{j[\phi_0 - \tau(\omega_0)(\omega-\omega_0)]}$$

$$= e^{j\phi_0} S(e^{j(\omega-\omega_0)}) e^{-j\tau(\omega_0)(\omega-\omega_0)} \xleftrightarrow{DTFT} y[n]$$

$$y[n] = e^{j\phi_0} s[n-\tau(\omega_0)] e^{j\omega_0 n}$$

Group delay $\tau(\omega) = -\frac{d}{d\omega} \arg\{H(e^{j\omega})\}$

1) Delay experienced by narrowband envelope of signal at ω

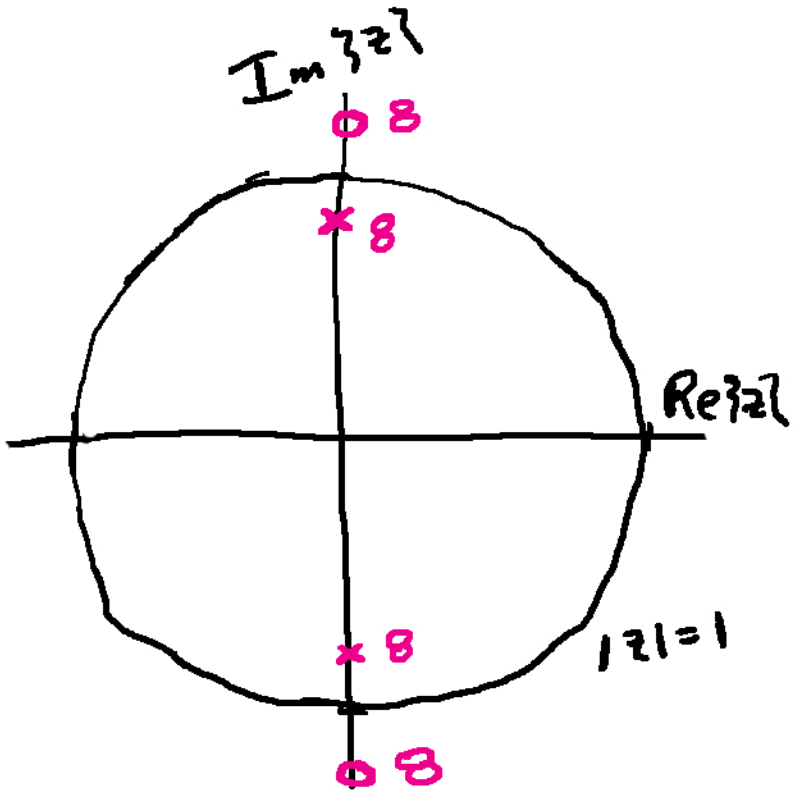
2) Linear phase $\longleftrightarrow$ constant group delay

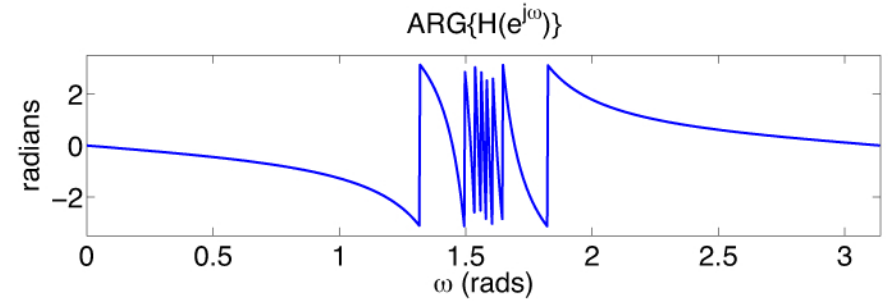
Nonlinear phase $\longleftrightarrow$ non constant group delay

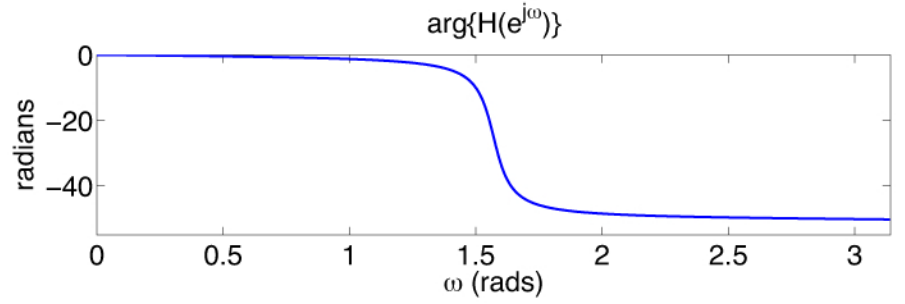
different frequency packets experience different delays

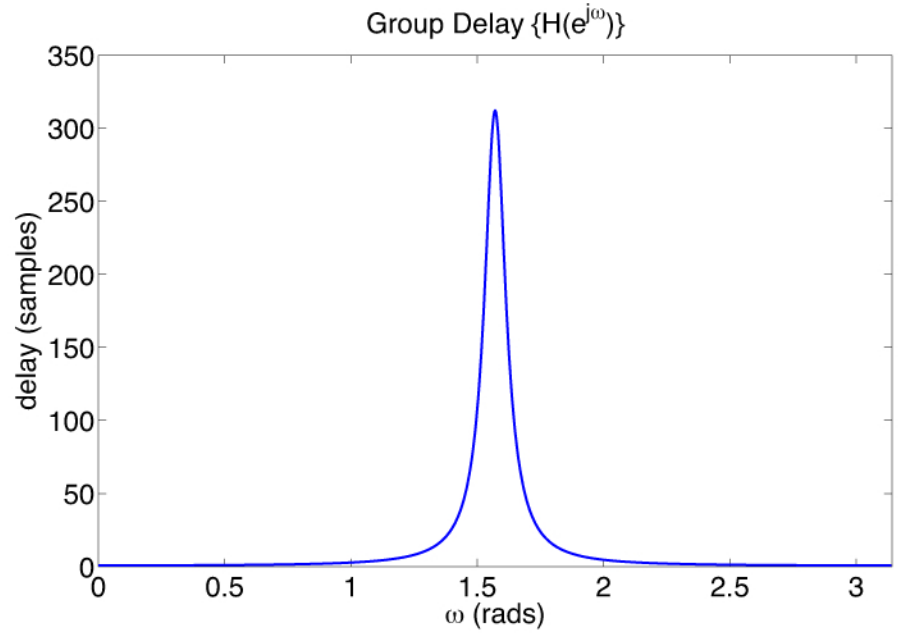
Example: Allpass

$$H(z) = \left(\frac{z^{-2} + .95^2}{1 + .95^2 z^{-2}} \right)^8$$



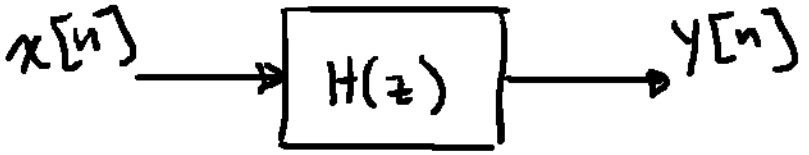


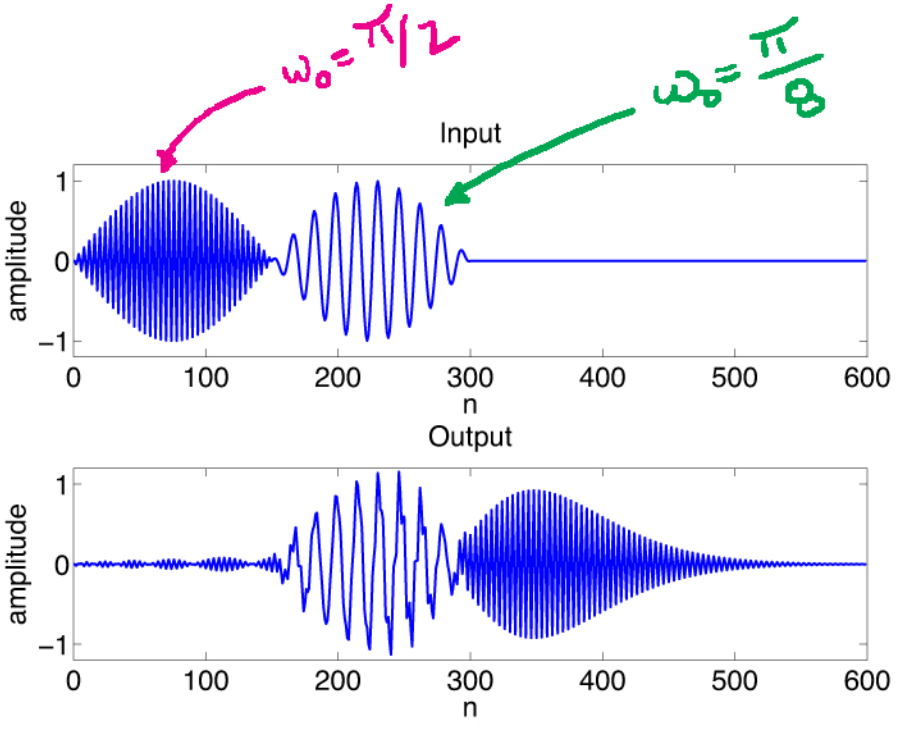


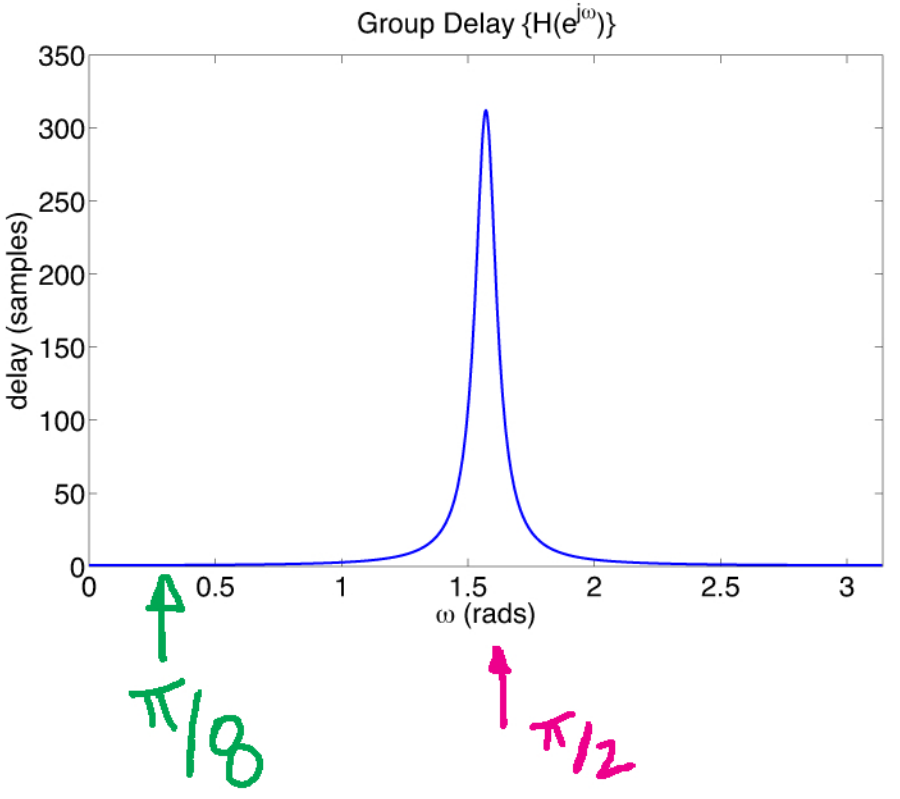


Example: Allpass

$$H(z) = \left(\frac{z^{-2} + .95^2}{1 + .95^2 z^{-2}} \right)^8$$







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